Learning how to prove Workshop Week 8

Groups

The Fundamental Theorem of Finite Cyclic Groups

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1. In each case determine all the subgroups of the given cyclic group and draw its lattice diagram.
   (a) $G = \langle g \rangle$ with $\mathrm{o}(g) = 8$
   (b) $\mathbb{Z}_{pq}$, where p and q are distinct primes.

2. (a) Find and describe (by listing all the elements) the subgroups of $\mathbb{Z}_{36}$.
   (b) Draw the lattice diagram of $\mathbb{Z}_{36}$.